

Gönül Demir

Senior Electronics / Hardware R&D Engineer

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gonuldemir.github.io

Engineering portfolio, articles & projects



TECHNICAL EXPERTISE

Optical Design | Sensor & Hardware Design | Electromagnetic Compatibility
Embedded Software | Production & Field Support | Testing & Documentation

Over 15 years of experience taking sensor and optoelectronic products from design through testing, certification and serial production. Manufacturing and field troubleshooting inform my focus on manufacturability, verifiable performance and reliability in real operating conditions.

PROFESSIONAL EXPERIENCE

Career Break | Professional Development & Technical Writing

Nov 2024 - Present

Improved my English and learned basic German (A1). Created my engineering blog, where I publish technical articles and document projects.

Teknim Electronic | Senior R&D Engineer

Istanbul, Jun 2022 - Nov 2024

Designed the optical arrangement and electronic hardware of a new beam smoke detector from the ground up. Demonstrated a detection range of **over 100 m in prototype testing**. Reduced bill-of-materials cost by **10%** through changes to optical and electronic component selection.

Fenac Encoders & Sensors | R&D Engineer

Istanbul, Apr 2017 - May 2022

Elevator Door Detection System

Designed the control board, optical arrangement, and power board for the FNC DD series elevator door detector.

Encoder & Sensor Electronics

Designed and verified electronic hardware for optical and magnetic encoders used in industrial and military applications, including schematic capture and PCB layout.

Developed both hardware and embedded software for diffuse photoelectric sensors. Also developed electronic hardware for time-of-flight (ToF) distance sensors and MEMS-based tilt measurement systems.

Production & Field Support

Supported product introduction into production and investigated manufacturing and field failures. Visited customer sites to diagnose problems under actual operating conditions. Designed test equipment and fixtures to improve the speed, accuracy, and repeatability of production testing.

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PROFESSIONAL EXPERIENCE - CONTINUED

Mavili Electric Electronic | R&D Engineer

Istanbul, Aug 2011 - Oct 2016

Detector Design

Designed the electronic hardware for all six models in a beam smoke detector family. For the separate transmitter-receiver models, developed the electronics around an existing optical and mechanical design. For the reflector-based models, designed both the optical arrangement and electronic hardware from the ground up.

Also designed the electronic hardware and developed the initial prototypes of flame detectors.

Testing & Certification

Prepared technical documentation, developed test procedures, and conducted tests for both products I designed and other products in the company's portfolio. Contributed to successful beam smoke detector certification through BRE/LPCB, TSE, and certification bodies in Bulgaria.

Played an active role in establishing the in-house EMC testing facility and designed dedicated test equipment and fixtures as needed.

Production & Product Improvement

Investigated manufacturing and field failures in products that had been in serial production for years. Contributed to design and process improvements that reduced the manufacturing-related failure rate from **7% to 2%** and shortened test times by **15%**.

Inovenso Consulting R&D Machinery | R&D Engineer - Freelance

Istanbul, Sep 2010 - Jun 2011

Developed motor-speed measurement circuits and motor-driver boards for the company's early textile machinery. As the project's sole electronics engineer, handled hardware design, embedded software, and electronic assembly.

Investigated board failures during high-speed, high-power motor operation, gaining practical experience in electromagnetic interference and its effects on circuit reliability.

Soypak Electronic | Purchasing Manager

Istanbul, Dec 2009 - Aug 2010

Sourced electronic components from suppliers in East Asia and handled technical correspondence with suppliers and customers. A motor-driver development project for a customer provided hands-on electronic design experience and prompted my transition into R&D.

For more details, visit my blog: gonuldemir.github.io/about/

EDUCATION

BSc, Electronics Engineering | Uludağ University | Sep 2004 - Jun 2009

Erasmus Exchange, Electronics Engineering | University of Salerno, Italy | 2008 - 2009

SEMINARS & TRAINING

Fresu Electronics - Dario Fresu: EMC/EMI Design Course | 2026

Würth Elektronik: Electromagnetic Compatibility Design & Power EMI Compliance Issues Seminar

Nail Turunç - Motivation Workshop: Leadership-Oriented Coaching | 30 hours | Nov 2024

LANGUAGES & TOOLS

Languages: Turkish: Native | English: Professional working proficiency | German: A1

Tools: Altium Designer, KiCad, LTspice/SPIICE, laboratory test instruments.

Embedded software: STM32 firmware development in C; basic experience with Visual Studio.